E18: Exact Lyapunov Descent for the Regularized WCT Curvature Energy

Proof note for Manuscript 2

October 26, 2026

Abstract

We prove a precise conditional version of E18: the regularized Wave Confinement Theory (WCT) curvature-plus-gradient energy is nonnegative and nonincreasing along its exact negative L^2 gradient flow. We compute the full complex variational derivative, including differentiation of the nonlinear curvature weight, and obtain an exact energy–dissipation identity. Periodic boundary conditions and classical solution regularity are imposed to make every integration by parts and chain rule legitimate. The theorem does not establish global existence of the flow or identify it with the independently proposed physical-time WCT dynamics.

1 Definition and assumptions

Let $\Omega = \mathbb{T}^d$ be a flat d-dimensional torus, $1 \le d \le 3$. Fix $c_1, c_2 \ge 0$, $\varepsilon > 0$, and $\alpha \in \mathbb{R}$. For a complex field $\psi : \Omega \to \mathbb{C}$ set

$$\rho = |\psi|^2, \tag{1}$$

$$D(\rho) = \rho + \varepsilon^2 e^{-2\alpha\rho}, \tag{2}$$

$$R_\varepsilon(\psi) = \frac{\overline{\psi}}{D(|\psi|^2)}, \tag{3}$$

$$\Theta_\varepsilon[\psi] = -(\Delta\psi) R_\varepsilon(\psi), \tag{4}$$

$$F(\rho) = \frac{\rho}{D(\rho)^2}. \tag{5}$$

The energy is

$$\boxed{\mathcal{E}[\psi] = \int_\Omega \left(c_1 |\nabla\psi|^2 + c_2 |\Theta_\varepsilon[\psi]|^2 \right) dx = \int_\Omega \left(c_1 |\nabla\psi|^2 + c_2 F(|\psi|^2) |\Delta\psi|^2 \right) dx.} \tag{6}$$

All variational derivatives below use the convention

$$\mathrm{D}\mathcal{E}(\psi)[h] = 2\,\mathrm{Re} \int_\Omega \frac{\delta \mathcal{E}}{\delta \overline{\psi}} \overline{h}\, dx, \tag{7}$$

where h is a complex variation and the underlying tangent space is real. In particular, $\delta\mathcal{E}/\delta\overline{\psi}$ is one-half of the real-L^2 Hilbert gradient for the inner product $(u, v)_{\mathbb{R}} = \mathrm{Re} \int_\Omega u\overline{v}\, dx$.

Proposition 1 (Well-defined, nonnegative energy)**.** *The denominator $D(\rho)$ is strictly positive for every $\rho \ge 0$. For every $\psi \in H^2(\Omega; \mathbb{C})$, the functional in (6) is finite and nonnegative.*

Proof. As $\varepsilon > 0$, one has $D(\rho) = \rho + \varepsilon^2 e^{-2\alpha\rho} > 0$. Thus F is smooth and nonnegative on $[0, \infty)$. For $d \le 3$, $H^2(\Omega)$ embeds continuously in $L^\infty(\Omega)$; therefore $F(|\psi|^2)$ is bounded, while $\nabla\psi, \Delta\psi \in L^2(\Omega)$. Both terms in (6) are finite and nonnegative because $c_1, c_2 \ge 0$. □

2 Exact first variation

The derivative of the curvature weight is

$$D'(\rho) = 1 - 2\alpha\varepsilon^2 e^{-2\alpha\rho}, \tag{8}$$

$$F'(\rho) = \frac{D(\rho) - 2\rho D'(\rho)}{D(\rho)^3} \tag{9}$$

$$= \boxed{\frac{\varepsilon^2 e^{-2\alpha\rho}(1 + 4\alpha\rho) - \rho}{\left(\rho + \varepsilon^2 e^{-2\alpha\rho}\right)^3}}. \tag{10}$$

Proposition 2 (Real Fréchet differentiability on H^2)**.** *For fixed $1 \le d \le 3$, $\varepsilon > 0$, $\alpha \in \mathbb{R}$, and real c_1, c_2, the energy (6) is a continuously Fréchet differentiable map*

$$\mathcal{E} : H^2(\mathbb{T}^d; \mathbb{C})_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Its derivative, taken over the real scalar field, is the bounded real-linear functional

$$\begin{aligned} \mathrm{D}\mathcal{E}(\psi)[h] = {} & 2c_1 \,\mathrm{Re} \int_\Omega \nabla\psi \cdot \nabla\overline{h}\, dx \\ & + 2c_2 \,\mathrm{Re} \int_\Omega F(|\psi|^2) \Delta\psi\, \Delta\overline{h}\, dx \\ & + 2c_2 \,\mathrm{Re} \int_\Omega F'(|\psi|^2) \psi |\Delta\psi|^2 \overline{h}\, dx. \end{aligned} \tag{11}$$

More precisely, for each $M < \infty$ there is $C_M < \infty$, depending on the fixed parameters and the torus, such that whenever $\|\psi\|_{H^2} \le M$ and $\|h\|_{H^2} \le 1$,

$$\left|\mathcal{E}[\psi + h] - \mathcal{E}[\psi] - \mathrm{D}\mathcal{E}(\psi)[h]\right| \le C_M \|h\|_{H^2}^2. \tag{12}$$

The derivative is locally Lipschitz in the operator norm of $\mathcal{L}_{\mathbb{R}}(H^2, \mathbb{R})$.

Proof. Write $\delta = \|h\|_{H^2}$ and use the embedding $H^2(\mathbb{T}^d) \hookrightarrow L^\infty$ for $d \le 3$. Let $q = \Delta\psi$, $r = \Delta h$, $\rho = |\psi|^2$, $\eta = \psi\overline{h} + \overline{\psi}h$, and $\zeta = |h|^2$. Then

$$|\psi + h|^2 = \rho + \eta + \zeta, \qquad |\Delta(\psi + h)|^2 = |q|^2 + 2\,\mathrm{Re}(q\overline{r}) + |r|^2.$$

On the prescribed bounded H^2 ball,

$$\|\eta\|_\infty \le C_M\delta, \quad \|\zeta\|_\infty \le C_M\delta^2, \quad \|r\|_2 \le \delta, \quad \|q\|_2 \le M.$$

Both ρ and $\rho+\eta+\zeta$ belong to a fixed compact subset of $[0, \infty)$; positivity of D implies that F, F', F'' are bounded there. Taylor's theorem on the segment between these nonnegative endpoints therefore gives

$$F(\rho + \eta + \zeta) = F(\rho) + F'(\rho)\eta + B, \qquad \|B\|_\infty \le C_M\delta^2.$$

The linear part of $F(\rho + \eta + \zeta)|q + r|^2 - F(\rho)|q|^2$ is

$$2F(\rho)\,\mathrm{Re}(q\overline{r}) + F'(\rho)\eta|q|^2.$$

After subtracting it, the remaining terms are exactly

$$F(\rho)|r|^2 + 2F'(\rho)\eta\,\mathrm{Re}(q\overline{r}) + F'(\rho)\eta|r|^2 + B|q|^2 + 2B\,\mathrm{Re}(q\overline{r}) + B|r|^2.$$

Their integrals are bounded by $C_M\delta^2$ using Cauchy–Schwarz, the displayed L^∞ bounds, and $\delta \le 1$. The gradient part expands exactly as

$$|\nabla(\psi+h)|^2 = |\nabla\psi|^2 + 2\,\mathrm{Re}(\nabla\psi\cdot\nabla\overline{h}) + |\nabla h|^2;$$

its remainder is likewise $O(\delta^2)$. Since $F'(\rho)\eta|q|^2 = 2\,\mathrm{Re}(F'(\rho)\psi|q|^2\overline{h})$, this proves both (11) and (12). Each term in (11) is bounded by $C_M\|h\|_{H^2}$, so the displayed formula is a real bounded linear functional.

For completeness, if ψ, φ lie in a bounded H^2 ball, the smooth scalar compositions satisfy

$$\|F(|\psi|^2) - F(|\varphi|^2)\|_\infty + \|F'(|\psi|^2)\psi - F'(|\varphi|^2)\varphi\|_\infty \le C_M\|\psi-\varphi\|_{H^2}.$$

Furthermore,

$$\big\||\Delta\psi|^2 - |\Delta\varphi|^2\big\|_1 \le (\|\Delta\psi\|_2 + \|\Delta\varphi\|_2)\|\Delta(\psi-\varphi)\|_2.$$

Insert these estimates into (11), using $\|h\|_\infty \le C\|h\|_{H^2}$, to obtain $\|\mathrm{D}\mathcal{E}(\psi) - \mathrm{D}\mathcal{E}(\varphi)\|_{\mathrm{op}} \le C_M\|\psi-\varphi\|_{H^2}$. Thus $\mathcal{E}$ is C^1 as a real Banach-space functional. □

Proposition 3 (Full complex variational derivative)**.** *For $\psi \in H^4(\Omega;\mathbb{C})$, the real Fréchet derivative of Proposition 2 has the following L^2 representative in convention (7):*

$$\boxed{G[\psi] := \frac{\delta\mathcal{E}}{\delta\overline{\psi}} = -c_1\Delta\psi + c_2\Delta\big(F(|\psi|^2)\Delta\psi\big) + c_2F'(|\psi|^2)\psi|\Delta\psi|^2.} \tag{13}$$

Proof. Let $h \in C^\infty(\Omega;\mathbb{C})$ and $\psi_s = \psi + sh$ for real s. Direct differentiation gives

$$\frac{d}{ds}|\psi_s|^2\Big|_{s=0} = \psi\overline{h} + \overline{\psi}h, \tag{14}$$

$$\frac{d}{ds}|\Delta\psi_s|^2\Big|_{s=0} = \Delta h\,\Delta\overline{\psi} + \Delta\psi\,\Delta\overline{h}. \tag{15}$$

For the gradient part, periodic integration by parts yields

$$\mathrm{D}\Big(c_1\|\nabla\psi\|_2^2\Big)[h] = 2c_1\,\mathrm{Re}\int_\Omega \nabla\psi\cdot\nabla\overline{h}\,dx = -2c_1\,\mathrm{Re}\int_\Omega \Delta\psi\,\overline{h}\,dx. \tag{16}$$

For the curvature part the product rule gives

$$\begin{aligned}&\mathrm{D}\bigg(c_2\int_\Omega F(|\psi|^2)|\Delta\psi|^2dx\bigg)[h]\\ &\quad= 2c_2\,\mathrm{Re}\int_\Omega F(|\psi|^2)\Delta\psi\,\Delta\overline{h}\,dx + 2c_2\,\mathrm{Re}\int_\Omega F'(|\psi|^2)\psi|\Delta\psi|^2\overline{h}\,dx.\end{aligned} \tag{17}$$

The Laplacian is self-adjoint on $\mathbb{T}^d$, so

$$\int_\Omega F(|\psi|^2)\Delta\psi\,\Delta\overline{h}\,dx = \int_\Omega \Delta\big(F(|\psi|^2)\Delta\psi\big)\overline{h}\,dx. \tag{18}$$

Combining this with (16) and (17),

$$\mathrm{D}\mathcal{E}(\psi)[h] = 2\,\mathrm{Re}\int_\Omega G[\psi]\overline{h}\,dx. \tag{19}$$

This is (13). More precisely, H^4 regularity and smooth composition imply $F(|\psi|^2) \in H^2$; also $\Delta\psi \in H^2$. Since H^2 is an algebra for $d \le 3$, $F(|\psi|^2)\Delta\psi \in H^2$, and its Laplacian belongs to L^2. The remaining nonlinear summand belongs to L^2 because $F'(|\psi|^2)\psi$ and $\Delta\psi$ are bounded and $\Delta\psi \in L^2$. The periodic integration-by-parts identities extend by density from smooth h to all $h \in H^2$. Thus the displayed representation holds for the full Fréchet derivative, not merely its smooth directional restrictions. □

3 Exact gradient-flow dissipation theorem

Theorem 1 (E18, conditional exact Lyapunov descent)**.** *Let $\psi \in C^1([0,T]; H^4(\mathbb{T}^d;\mathbb{C}))$ solve the specified negative-gradient evolution*

$$\boxed{\partial_t\psi = -G[\psi] = c_1\Delta\psi - c_2\Delta(F(|\psi|^2)\Delta\psi) - c_2F'(|\psi|^2)\psi|\Delta\psi|^2.} \tag{20}$$

Then, for every $t \in [0,T]$,

$$\boxed{\frac{d}{dt}\mathcal{E}[\psi(t)] = -2\|G[\psi(t)]\|^2_{L^2(\Omega)} = -2\|\partial_t\psi(t)\|^2_{L^2(\Omega)} \le 0.} \tag{21}$$

For any $0 \le t_1 \le t_2 \le T$,

$$\boxed{\mathcal{E}[\psi(t_2)] + 2\int_{t_1}^{t_2} \|G[\psi(s)]\|_2^2\, ds = \mathcal{E}[\psi(t_1)].} \tag{22}$$

In particular, $\mathcal{E}[\psi(t_2)] \le \mathcal{E}[\psi(t_1)]$.

Proof. The continuous embedding $H^4 \hookrightarrow H^2$ makes the trajectory C^1 as an H^2-valued map. Proposition 2 proves that $\mathcal{E} : H^2_{\mathbb{R}} \to \mathbb{R}$ is C^1; hence the genuine real Banach-space chain rule, followed by Proposition 3, gives

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = \mathrm{D}\mathcal{E}(\psi(t))[\partial_t\psi(t)] \tag{23}$$

$$= 2\,\mathrm{Re}\int_\Omega G[\psi(t)]\,\overline{\partial_t\psi(t)}\,dx. \tag{24}$$

Inserting $\partial_t\psi = -G[\psi]$ yields

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = -2\,\mathrm{Re}\int_\Omega G[\psi(t)]\overline{G[\psi(t)]}\,dx \tag{25}$$

$$= -2\int_\Omega |G[\psi(t)]|^2 dx \le 0, \tag{26}$$

proving (21). Integrating over $[t_1,t_2]$ proves (22), and dropping the nonnegative dissipation integral gives monotonicity. □

Corollary 1 (Characterization of zero instantaneous dissipation)**.** *Under the hypotheses of Theorem 1, at each t one has*

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = 0 \quad\Longleftrightarrow\quad G[\psi(t)] = 0 \text{ in } L^2 \quad\Longleftrightarrow\quad \partial_t\psi(t) = 0. \tag{27}$$

Thus a nonstationary time slice has strictly negative energy derivative.

Proof. The equivalent conditions follow immediately from (21), the nonnegativity of a squared norm, and the evolution equation (20). □

Corollary 2 (Finite accumulated dissipation)**.** *If the hypotheses of Theorem 1 hold on every finite interval $[0,T]$ and the solution exists globally for $t \ge 0$, then*

$$\int_0^\infty \|G[\psi(t)]\|_2^2 dt \le \frac{1}{2}\mathcal{E}[\psi(0)]. \tag{28}$$

Proof. By Proposition 1 and (22), $2\int_0^T \|G[\psi(t)]\|_2^2 dt \le \mathcal{E}[\psi(0)]$ for every T. Monotone convergence of the nonnegative time integral proves (28). □

4 Boundary terms and conventions

Remark 1 (Nonperiodic domains)**.** *For a sufficiently smooth bounded domain, the first variation has, in addition to the bulk term in* (13)*, the boundary contribution*

$$B[h] = 2\,\mathrm{Re}\int_{\partial\Omega}[(c_1\partial_n\psi - c_2\partial_n(F(|\psi|^2)\Delta\psi))\overline{h} + c_2F(|\psi|^2)\Delta\psi\,\partial_n\overline{h}]\,dS. \tag{29}$$

The bulk L^2-gradient derivation applies when the admissible variations and boundary conditions make $B[h] = 0$, and the evolution is compatible with those boundary conditions. Imposing an unspecified boundary condition is not enough for a fourth-order integration-by-parts argument.

Remark 2 (Factor of two)**.** *With the convention* (7)*, the exact flow $\psi_t = -\delta\mathcal{E}/\delta\overline{\psi}$ yields the coefficient* 2 *in* (21)*. With the real Hilbert gradient $\nabla_{\mathbb{R}}\mathcal{E} = 2G$, the flow $\psi_t = -\nabla_{\mathbb{R}}\mathcal{E}$ instead obeys $d\mathcal{E}/dt = -\|\nabla_{\mathbb{R}}\mathcal{E}\|_2^2$. The factor is a normalization choice, not a substantive discrepancy.*

5 Scope and limitations

Theorem 1 establishes an *exact conditional* Lyapunov identity for solutions of *the specified gradient flow*. It does not assert: (i) local or global existence for arbitrary initial data; (ii) uniform parabolicity, since $F(0) = 0$; (iii) convergence to a unique equilibrium; (iv) whole-space spatial localization; or (v) derivation of (20) from a distinct Hamiltonian, wave, or mixed dissipative WCT dynamical equation.

For an arbitrary specified evolution $\psi_t = X[\psi]$, the same chain rule gives only

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = 2\,\mathrm{Re}\int_\Omega G[\psi]\,\overline{X[\psi]}\,dx, \tag{30}$$

and this expression does not have a fixed sign without additional compatibility assumptions.
Conclusion. The mathematically precise E18 statement is

$$\boxed{\psi_t = -\frac{\delta\mathcal{E}}{\delta\overline{\psi}} \quad\Longrightarrow\quad \frac{d\mathcal{E}}{dt} = -2\left\|\frac{\delta\mathcal{E}}{\delta\overline{\psi}}\right\|_2^2 \le 0,}$$

for the corrected regularized energy, on the stated class of solutions.

References

[1] R. J. Reyes, *WCT Full Equation List Corrected*, E17–E18, online repository, https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md.

[2] R. J. Reyes, *WCT Lean Derived Audit*, E18 conditional algebraic descent core, https://github.com/rickyjreyes/wct-lean/blob/main/WCTLean/DerivedAudit.lean.